

The motivation of the paper is to present a systematic study on Google’s Web Light service. The median mobile page size has increased by 300% over the years, and to improve the browsing experience on slow mobile clients, Google introduced Web Light service in 2015. The service works seamlessly with the Android and Chrome browsers on slow mobile clients and can help with improving user perceived performance, saving user data and increasing traffic to webpages. The study examines how Web Light modifies webpages, investigates its impact on advertisements, quantifies page load performance and highlights important privacy concerns. Web Light can also help circumvent censorship.

The weblight service transcodes webpages by either removing images, reducing image resolutions, removing javascript, changing CSS or replacing videos with images. The paper’s results show up to 15x fewer images and 12x smaller pages. WebLight also reduces the number of ads, showing up to three ads on a page and improves median PLT by 5.4x. It can provide fast access to censored content - outperforming Tor and common VPNs, however, It does not preserve end to end encryption.

On slow devices in supported countries, Web Light works seamlessly, however, it does not transcode webpages that require login and some video streaming websites. It also limits the number of ads to the first three ads requested by the webpage. The service does provide an opt out option for publishers. The authors studied websites from Alexa top 5000 and found that Web Light transcodes $> 70\%$ of them. They also constructed web pages of various sizes to determine the thresholds of content amount and sizes beyond which Web Light starts transcoding. The authors also analyze the impact of caching.

The experimental results show that Web Light only transcodes pages if the base HTML size is greater than 150 bytes, however it embeds its own scripts and elements which are between 10KB and 12KB in size. Image transcoding is also triggered only when the base web page is above the 150 bytes threshold. When they images are transcoded, their width is reduced to 300 pixels and height is adjusted accordingly. By default, Web Light removes all scripts, except for scripts within iframes used for fetching ads and also removes all images fetched through javascript. CSS is also converted to inline. Web Light only supports a subset of ad networks and limits the number of ads to 3, which can impact the ad revenue. Web Light can also help bypass server side geo-blocking. The median PLT is reduced from 19s to 6.5s and median speed index is also reduced by 3.3x. Caching by Web Light also reduced the median PLT by 3.8x. Web Light also provides fast access to censored content.

The study sheds light on the benefits and potential harms of the Google’s Web Light service. While it makes the web more accessible, it can harm the content publishers and poses privacy risks for the users. An interesting future work of this study would be to separately analyze client side rendered, server side rendered and server side generated web pages. Libraries like ReactJS allow the developers to create interactive websites purely through JavaScript and the base HTML footprint may be very small. Likewise, frameworks like NextJS offer server side generation and rendering of web pages, allowing them to load faster on the client side. It would be interesting to see how it affects such pages.